

ans 3 $\alpha \in \mathbb{V}^7$ dBT

$$* (f + \lambda g)^{(0)} = f + \lambda g = f^{(0)} + \lambda g^{(0)}$$

$$(f + \lambda g)^{(n+1)} = ((f + \lambda g)^{(n)})' = (f^{(n)} + \lambda g^{(n)})' = f^{(n+1)} + \lambda g^{(n+1)}$$

$$* (fg)^{(0)} = f^{(0)}g^{(0)} = f^{(0)}g^{(0)}$$

$$(fg)^{(n+1)} = ((fg)^{(n)})' = \left(\sum_{k=0}^n C_n^k f^{(k)} g^{(n-k)} \right)' = \sum_{k=0}^n C_n^k f^{(k+1)} g^{(n-k)} + \sum_{k=0}^n C_n^k f^{(k)} g^{(n-k+1)}$$

$$* (fg)^{(n)} = f^{(n)}g + f'g' = \sum_{k=0}^n C_n^k f^{(k)} g^{(n-k)} = \sum_{k=0}^n C_n^k f^{(n-k+1)} g^{(k)}$$

en exercice

$$\begin{aligned}
 (fg)^{(n+1)} &= \cancel{f^{(n+1)}} \cancel{g} + \sum_{k=0}^{n+1} \cancel{C_n^k} \\
 &= f^{(n+1)} g + \sum_{k=1}^n C_n^k f^{(n-k+1)} g^{(k)} + \sum_{k=0}^{n-1} C_n^k f^{(n-k)} g^{(k+1)} + f^{(n)} g^{(n+1)}
 \end{aligned}$$

$$= f^{(n+1)} g + \sum_{k=1}^n C_n^k f^{(n-k+1)} g^{(k)} + \sum_{k=1}^n C_n^{k-1} f^{(n-k+1)} g^{(k)} + f^{(n)} g^{(n+1)}$$

$$= f^{(n+1)} g + \sum_{k=1}^n \underbrace{(C_n^k + C_n^{k-1})}_{C_{n+1}^k} f^{(n-k+1)} g^{(k)} + f^{(n)} g^{(n+1)}$$

$$= \sum_{k=1}^{n+1} C_{n+1}^k f^{(n+1-k)} g^{(k)}$$